LAB 6: DYNAMIC ROUTING CONFIGURATION: RIP, EIGRP, OSPF, BGP

OBJECTIVES:

- To study the fundamentals and working principles of dynamic routing protocols such as RIP, EIGRP, OSPF, and BGP.
- To configure and verify different dynamic routing protocols on routers using Cisco Packet Tracer for automatic route discovery.
- To analyze packet transmission between different networks using dynamic routing.

THEORY:

Dynamic routing is a networking technique in which routers automatically exchange routing information with other routers to determine the best path for data transmission. Unlike static routing, dynamic routing protocols adapt automatically to changes in network topology such as link failures or new networks. These protocols reduce manual configuration and improve scalability, reliability, and efficiency in modern networks.

Dynamic routing protocols are broadly classified based on their algorithms and areas of operation. In this experiment, four important dynamic routing protocols—RIP, EIGRP, OSPF, and BGP—are studied and configured.

1. RIP (Routing Information Protocol):

RIP is one of the oldest dynamic routing protocols and follows the distance-vector routing algorithm. It uses hop count as its routing metric to determine the best path to a destination network.

Operation:

RIP routers periodically broadcast their entire routing table to directly connected neighbors every 30 seconds. The maximum allowable hop count is 15; a hop count of 16 indicates that the network is unreachable. RIP version 2 (RIPv2) supports classless routing and subnet masks.

Applications:

RIP is mainly used in small networks where simplicity is more important than performance. Due to slow convergence and limited scalability, it is not suitable for large or complex networks.

2. EIGRP (Enhanced Interior Gateway Routing Protocol):

EIGRP is an advanced distance-vector routing protocol developed by Cisco. It is often referred to as a hybrid routing protocol because it combines features of both distance-vector and link-state routing.

Operation:
EIGRP uses the Diffusing Update Algorithm (DUAL) to calculate the most efficient path. It considers multiple metrics such as bandwidth, delay, reliability, and load. Unlike RIP, EIGRP sends only partial updates when changes occur, resulting in faster convergence and reduced bandwidth usage.

Applications:
EIGRP is widely used in Cisco-based enterprise networks where fast convergence, scalability, and efficient use of network resources are required.

3. **OSPF (Open Shortest Path First):**

OSPF is a link-state routing protocol designed for large and complex networks. It maintains a complete map of the network topology and calculates the shortest path to each destination.

Operation:
OSPF uses the Dijkstra Shortest Path First (SPF) algorithm to compute routes. Routers exchange Link State Advertisements (LSAs) to build a link-state database. OSPF supports hierarchical network design by dividing the network into areas, with Area 0 serving as the backbone.

Applications:
OSPF is commonly used in large enterprise and service provider networks due to its fast convergence, scalability, and support for hierarchical routing.

4. BGP (Border Gateway Protocol):

BGP is a path-vector routing protocol used for routing between different Autonomous Systems (AS). It is the primary routing protocol of the Internet.

Operation:
BGP makes routing decisions based on attributes such as AS-path, next-hop, and routing policies defined by network administrators. Unlike interior routing protocols, BGP focuses on policy-based routing rather than shortest path selection.

Applications:
BGP is used by Internet Service Providers (ISPs) and large organizations to manage routing between different networks and autonomous systems across the Internet.

NETWORK TOPOLOGY:

A network topology was created using two routers: Router2 (1841) and Router1 Sudip (1841), interconnected via a serial/WAN link using their serial interfaces. The topology consists of two separate Local Area Networks (LANs) connected through these routers.

Switch0 (2960-24TT) is connected to Router2 and provides connectivity to two end devices: PC0 (PC-PT) and PC1 (PC-PT), forming Network 192.168.1.0/24. Switch1 (2960-24TT) is connected to Router1 Sudip and provides connectivity to two end devices: PC2 (PC-PT) and PC3 (PC-PT), forming Network 192.168.2.0/24.

The serial connection between Router2 and Router1 Sudip enables inter-network communication, allowing devices in the 192.168.1.0/24 network to communicate with devices in the 192.168.2.0/24 network through dynamic routing protocols. Green indicators on the connections represent active and functioning network links in the Cisco Packet Tracer simulation environment.

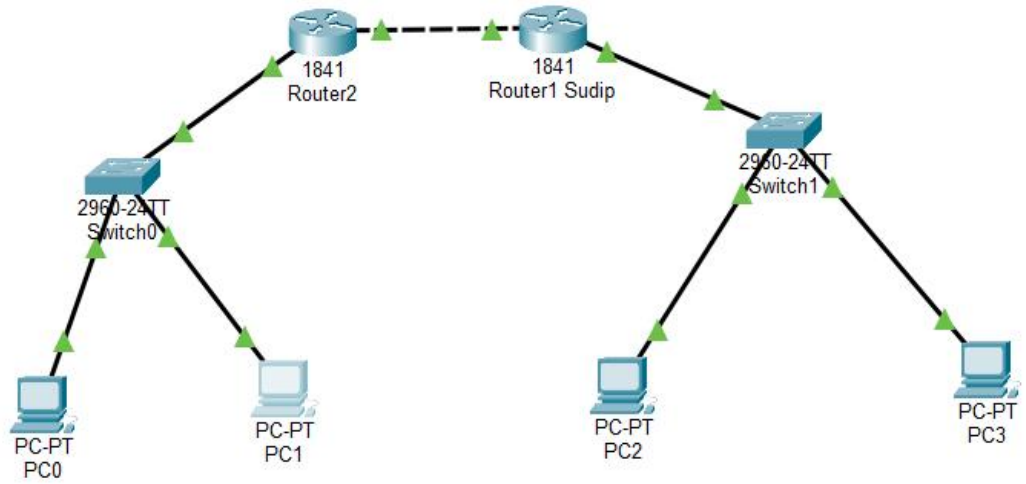

Fig: Network Topology

Configuration Table:

Network devices were configured with IP addresses and subnet masks as detailed in the table below:

IP Configuration Table:

Device	**Interface**	**IP Address**	**Subnet Mask**	**Default Gateway**
Router2	Gig0/1	192.168.1.1	255.255.255.0	N/A
Router2	Gig0/0	10.0.0.1	255.0.0.0	N/A
Router1 Sudip	Gig0/1	192.168.2.1	255.255.255.0	N/A
Router1 Sudip	Gig0/0	10.0.0.2	255.0.0.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

1. RIP Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Note: If you just see “Router >” instead of “Router(config)#” (as in the first line), then execute the following commands also.

```
Router>enable
Router# configure terminal
Router(config)#
```

Router1 (Right sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration Commands :

Router2 (Left sided one in topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
Router1 (Right sided one in topology):
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3.OSPF Configuration Commands:

```
Router2 (Left sided one in topology):
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0Router1
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands :

```
Router2 (Left sided one in topology) (AS 65001):
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
Router1 (Right sided one in topology) (AS 65002):
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

RESULT

The dynamic routing protocols were successfully implemented and tested across the network topology. Connectivity verification was performed by sending ping requests from PC2 (192.168.2.2) in the 192.168.2.0/24 network to PC0 (192.168.1.2) in the 192.168.1.0/24 network. Each routing protocol (RIP, EIGRP, OSPF, and BGP) demonstrated successful automatic route propagation and routing table updates. All packets were transmitted and received without loss, confirming proper inter-network communication and validating the correct configuration of the dynamic routing protocols.

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig: Ping (BGP configuration)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=2ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>
```

Fig: Ping (EIGRP)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Fig: Ping (OSPF)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Fig: Ping (RIP configuration)

DISCUSSION:

The laboratory experiment effectively demonstrated the configuration and operation of four dynamic routing protocols using Cisco Packet Tracer. RIP was the easiest protocol to configure, but it showed slower convergence because it relies on periodic routing updates. EIGRP provided faster convergence by using triggered updates and optimized bandwidth usage. OSPF required a more detailed setup, including area configuration, but it offered the advantage of being a vendor-neutral standard protocol. BGP involved the most complex configuration due to the need for autonomous system numbers and advanced routing policies. Connectivity testing using ping between PC1 and PC3 confirmed that all protocols successfully enabled automatic route exchange and communication between different networks.

CONCLUSION:

This laboratory task successfully met its objective of studying and configuring dynamic routing protocols on Cisco routers. RIP, EIGRP, OSPF, and BGP were implemented correctly and tested for proper functionality. The results verified that each protocol supports automatic route learning and dynamic updates without requiring manual route configuration. Overall, the experiment enhanced understanding of routing behavior, protocol differences, and practical network configuration skills.